INCH-POUND

MIL-B-49430/1F(ER)
1 August 1991
SUPERSEDING
MIL-B-49430/1E(ER)
26 April 1988

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5599/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

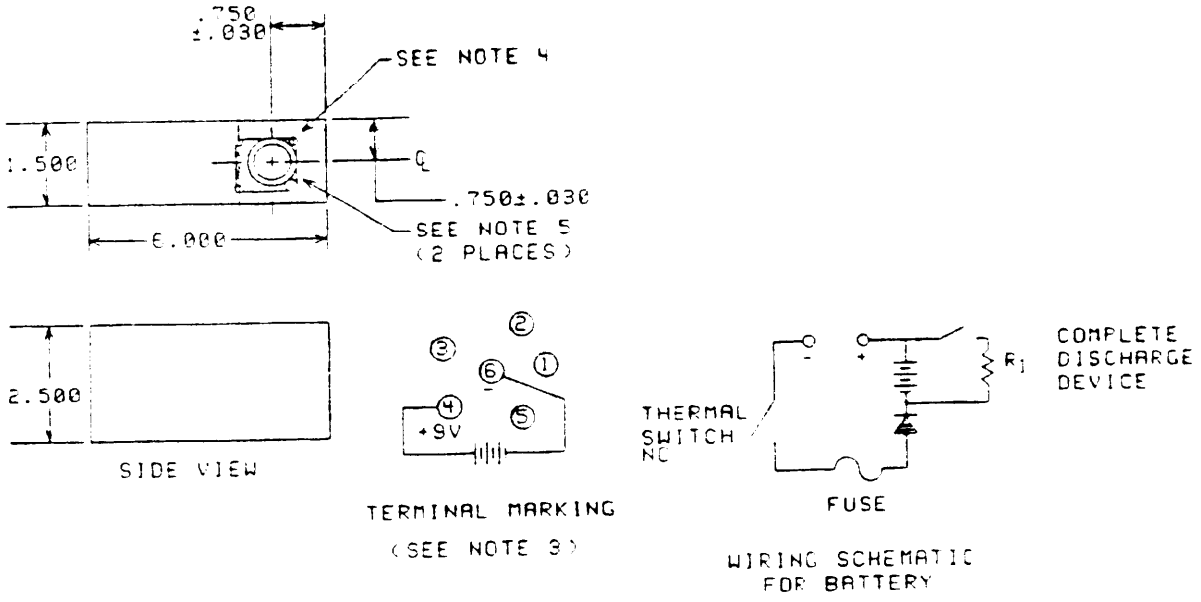


Figure 1. Battery dimensions and wiring schematic.

FSC 6135

AMSC N/A
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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are -.063, +.000.

3. Terminal marking to be printed or embossed on jackets in suitable location near the socket on the terminal face of the battery.

4. The battery terminal shall consist of a miniature circular battery connector, Dwg SC-C-179492 or equal. Connector shall be mounted in accordance with referenced drawing.

5. Connector socket body configuration shall accept two (2) each 0.156 inch diameter guideposts as shown. Depth of cavity shall be 0.657 inch minimum.

6. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

7. R_1 value shall be between 90 ohms and 125 ohms and have a power rating of 1.0 watt.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 9.15 volts.

Minimum voltage for capacity: 6.00 volts

The battery shall contain a 2.25 ampere non-replaceable slow blow fuse in the negative leg.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of the basic specification.

Terminals: Six (6) hole socket type with no obstructions of any holes (see note 5).

Weight (maximum): 1.0 pound (454 grams)

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 6.00 Volts
I	3.0
L	1.6
H	5.6
HT	5.4
LT	1.4
T	2.8
D	2.8

Initial voltage delay: During the I, H, HT, T, and D discharge tests, initial voltages below 6.0 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests initial voltages below 5.70 volts cannot exceed a 60.0 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraph 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

Connector: The connector shall be in accordance with SC-C-179492 with the following exception: The durability test described in paragraph 4h shall be performed 100 times with SM-D-687888. Each of the metallic sockets in the connector (top view) shall conform to Figure 2. Each socket shall be cylindrical in shape and may have a single gap with a maximum opening of 30 degrees. The socket shall accept a number 16 pin. Compliance with this requirement shall be demonstrated as a part of incoming inspection.

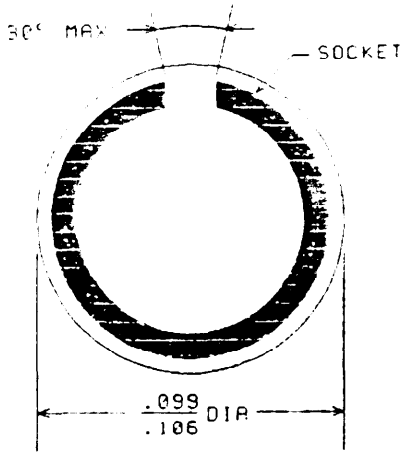


Figure 2. Socket (Top View)

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METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage: Details on storage conditions for all capacity tests are specified in the basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 2.0 amperes constant current at 70±5°F for a minimum of 2.7 hours to 2.00 volts. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 5.4 hours into voltage reversal.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage test: Load resistance of 8 ohms shall be used. Voltage shall be above 6.00 volts within 5.0 seconds.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Army-ER

Preparing Activity:
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